Progress in science proceeds in fits and starts. Some periods are filled with great breakthroughs; at other times researchers experience dry spells. Scientists put forward results, both theoretical and experimental. The results are debated by the community, sometimes they are discarded, sometimes they are modified, and sometimes they provide inspirational jumping-off points for new and more accurate ways of understanding the physical universe. In other words, science proceeds along a zig-zag path toward what we hope will be ultimate truth, a path that began with humanity's earliest attempts to fathom the cosmos and whose end we cannot predict. Whether string theory is an incidental rest stop along this path, a landmark turning point, or in fact the final destination we do not know. But the last two decades of research by hundreds of dedicated physicists and mathematicians from numerous countries have given us well-founded hope that we are on the right and possibly final track.

It is a telling testament of the rich and far-reaching nature of string theory that even our present level of understanding has allowed us to gain striking new insights into the workings of the universe. A central thread in what follows will be those developments that carry forward the revolution in our understanding of space and time initiated by Einstein's special and general theories of relativity. We will see that if string theory is correct, the fabric of our universe has properties that would likely have dazzled even Einstein.

Part II: The Dilemma of Space, Time, and the Quanta